

<div align="center">

■ SAMPARK

AI Driven Criminal Network Analysis System

![SIH 2026](https://sih.gov.in)

![FastAPI](https://fastapi.tiangolo.com)

![Firebase](https://firebase.google.com)

![Vercel](https://vercel.com)

![Python](https://python.org)

![License](LICENSE)

<p align="center">

Next-Generation AI-Driven Enterprise Intelligence Platform for Law Enforcement, Forensic Analysts, and Cyber Crime Investigation Units.

</p>

<p align="center">

■ Live Cloud Workstation •

■ GitHub Repository •

■■ System Architecture

</p>

</div>

■ Table of Contents

- ■ System Overview
- ■ Key Intelligence Capabilities
- ■■ High-Level Architecture
- ■ Responsive UI & Workstation Themes
- ■ Data Model & Person-Scoped Isolation
- ■■ Tech Stack
- ■ Quick Start & Installation
- ■ Firebase Cloud Database Setup
- ■ Live Deployment Links

■ System Overview

TRACE FINDERS is a state-of-the-art criminal network analysis workstation engineered for law enforcement agencies, cybercrime investigation cells, and forensic intelligence teams under Smart India Hackathon (SIH 2026 - Problem Statement SIH26189).

Traditional investigative tools suffer from context bleeding, static visualizations, and fragmented evidence data across isolated silos (CDR logs, bank wire ledgers, CCTV DVR streams, blockchain wallets, and social media OSINT). TRACE FINDERS resolves this by establishing a 100% Person-Scoped Investigation Data Engine, correlated through Multi-Hop Cross-Domain Evidence Fusion and Explainable AI (XAI) decision support.

■ Key Intelligence Capabilities

■ 1. 100% Person-Scoped Data Model & Dynamic Switching

- Zero Context Bleeding: Every primary suspect, associate, and person of interest maintains a strictly isolated investigative record tree.
- Dynamic Context Controller: Switching the active target subject instantly transforms all workstation panels, financial ledgers, communication trees, DVR video clips, timeline chronology, and graph networks.
- 6 Distinct Target Profiles Built-In:
 - P-001 Arjun Sharma — Primary Subject *(Logistics & Financial Controller)*
 - P-002 Rohan Mehta — Business Contact *(Cross-Border Importer)*
 - P-003 Priya Joshi — Associate *(Corporate Accounts Executive)*
 - P-004 Vikram Patil — Person of Interest *(Hawala Transactor)*
 - P-005 Neha Kulkarni — Employee *(Internal Transport Dispatcher)*
 - P-006 Arjun S. — Ambiguous Match Candidate *(Entity Resolution Pipeline)*

■■ 2. Interactive Vis.js Link Analysis Knowledge Graph

- 360° Drag & Pan Control: Fully interactive graph canvas supporting pinch zoom, unconstrained dragging, free node repositioning, and double-click target focus.
- 4 Algorithmic Layout Engines:
 - Tree (Root Top) — Directed hierarchical top-down flow centered on the active target subject.
 - Radial — Concentric multi-hop degree rings.
 - Hierarchy — Left-to-right temporal/role dependency layout.
 - Compact — High-density force-directed cluster layout.
- Floating Controls Overlay: Quick-action floating toolbar ([+] Zoom In, [–] Zoom Out, [■] Fit to Screen, [■] Reset Layout, [■] Re-Center Root Node).

■ 3. CCTV / DVR Forensics Workspace

- All-Camera Responsive Inventory Grid: Displays all 12 monitored camera locations (CAM-01 through CAM-12) simultaneously with zero clipped cards or hidden carousels.
- 24-Hour Recording Event Timeline: Visual marker timeline (00:00 to 23:59 IST) displaying exact timestamps of suspect sightings, vehicle ANPR triggers, and meeting events.
- 3-Column Split Analytics Layout:
 - Left Column: Chronological Evidence Timeline.
 - Center Column: Multi-Camera Video Clip Player with playback controls (■, ■ 10s, ■ 10s) & SHA-256 evidence reference tags.
 - Right Column: Cross-domain Related Entities & Camera Registry.

■ 4. Multi-Hop Evidence Fusion & Explainable AI (XAI)

- Cross-Domain Correlation Pipeline: Automatically links entity nodes across 5 analytical domains:
$$\text{Person Profile} \rightarrow \text{CDR} \rightarrow \text{Telecom Node} \rightarrow \text{ANPR} \rightarrow \text{CCTV Event} \rightarrow \text{Wire} \rightarrow \text{Financial Ledger} \rightarrow \text{Wallet} \rightarrow \text{Blockchain}$$
- XAI Assessment Engine: Generates human-understandable correlation rationale, confidence scores, flagging triggers, and alternative hypotheses to support judicial review.

■ 5. Official Investigation Dossier & A4 PDF Generator

- Embedded Profile Photo: Profile card displaying target portrait, aliases, identifier grid, and legal risk classification.
- Native Print & PDF Export: Instant A4-formatted PDF generation using @page { size: A4; margin: 15mm; } via browser print engine integration.

■■ High-Level Architecture

```
graph TD
    A[Investigator Browser / Mobile Client] -->|HTTPS / REST API| B[Vercel Serverless Gateway / FastAPI]
    B -->|Fail-Safe Cloud Sync| C[Firestore Realtime Database]
    B -->|Graph Engine| D[Vis.js Link Analysis Network]
    B -->|XAI Correlation| E[Multi-Hop Fusion Engine]
    B -->|Report Service| F[A4 PDF Dossier Generator]

    subgraph Firestore Cloud Datastore
        C --> C1[profiles: P-001 ... P-006]
        C --> C2[cases: TRX-2026-017]
        C --> C3[cctv_inventory: CAM-01 ... CAM-12]
        C --> C4[multi_hop_fusion & XAI]
        C --> C5[audit_logs & timeline]
    end
end
```

■ Responsive UI & Workstation Themes

- Pure OLED Pitch Black & Obsidian Theme: Engineered with #000000 pitch black background, #090a0f matte charcoal cards, #040406 obsidian headers, and #38bdf8 electric cyan accents for low eye strain during overnight operations.
- Mobile & Tablet Drawer Navigation: On screens \$le 768\text{px}\$, the sidebar smoothly transforms into a slide-out drawer (z-index: 2000) with touch-optimized scroll wrappers for all data tables.

■ Data Model & Person-Scoped Isolation

```
{
  "datastore": {
    "profiles": {
      "P-001": {
        "id": "P-001",
        "name": "Arjun Sharma",
        "role": "Primary Subject",
        "phone": "+91 98765 1201",
        "email": "arjun.sharma.demo@example.test",
        "vehicle": "MH12 AB 4821",
        "account_number": "XXXX4821",
        "wallet_address": "0xDEMO...A721"
      },
      "P-002": {
        "id": "P-002",
        "name": "Rohan Mehta",
        "role": "Business Contact",
        "phone": "+91 97765 4876",
        "email": "rohan.mehta.demo@example.test",
        "vehicle": "MH01 CR 7814",
        "account_number": "XXXX7312",
        "wallet_address": "0xDEMO...D492"
      }
    }
  }
}
```

■■ Tech Stack

- | Layer | Technology / Library |
- | :--- | :--- |
- | Frontend UI | HTML5, CSS3 Variables, ES6 JavaScript, Vis.js Network Engine |
- | Backend REST API | Python 3.12, FastAPI, Uvicorn |
- | Cloud Database | Firebase Realtime Database & Firestore REST Service |
- | Serverless Deployment | Vercel Serverless Functions (@vercel/python) |
- | Source Control | Git, GitHub Actions / CI |

■ Quick Start & Installation

1. Clone the Repository

```
git clone https://github.com/nikhil24x5183-netizen/tracefinders.git
cd tracefinders
```

2. Install Dependencies

```
pip install -r requirements.txt
```

3. Run Local Development Server

```
python -m uvicorn backend.app:app --host 0.0.0.0 --port 8000
```

Open <http://localhost:8000> in your browser.

■ Firebase Cloud Database Setup

To link your personal Firebase Console project:

1. Open <https://console.firebase.google.com>.
2. Select your project (e.g. tracefinders-daa12).
3. Click Build → Realtime Database → Create Database (Start in Test Mode).
4. Set environment variable on Vercel or locally:

```
bash
```

```
export FIREBASE_DATABASE_URL="https://your-project-id-default-rtdb.firebaseio.com"
```

■ Live Deployment Links

- ■ Vercel Live Production App: <https://criminalnetworkanalysis.vercel.app>
 - ■ GitHub Repository: <https://github.com/nikhil24x5183-netizen/tracefinders>
 - ■ Firebase JSON Feed: <https://tracefinders-daa12-default-rtdb.firebaseio.com/datastore.json>
 - ■■ Architecture Documentation: ARCHITECTURE.md
-

■ License & Acknowledgments

Distributed under the MIT License. Developed for Smart India Hackathon 2026 (Problem SIH26189).